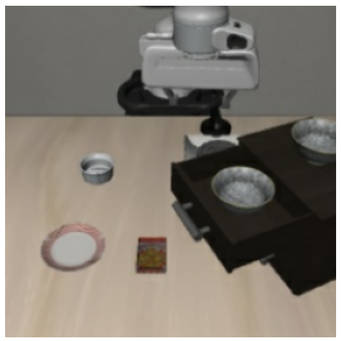
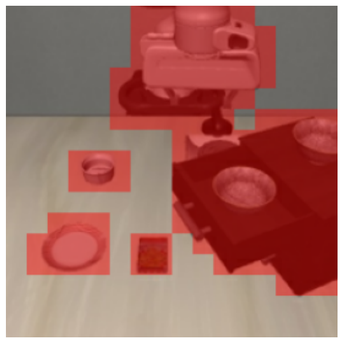
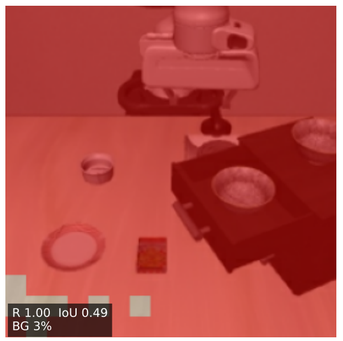
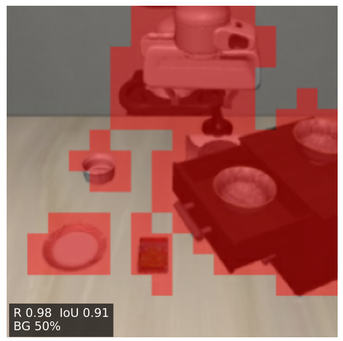
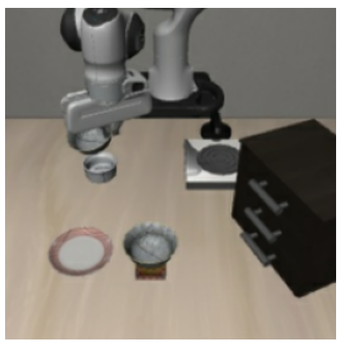
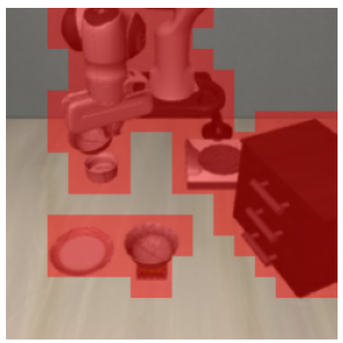
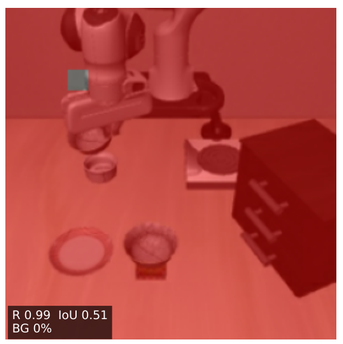
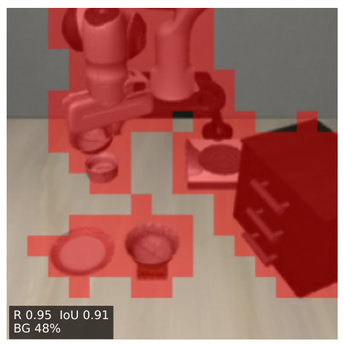
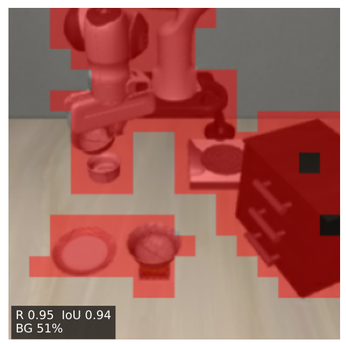
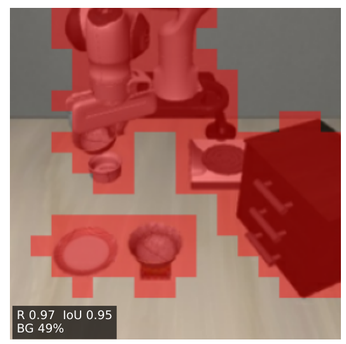
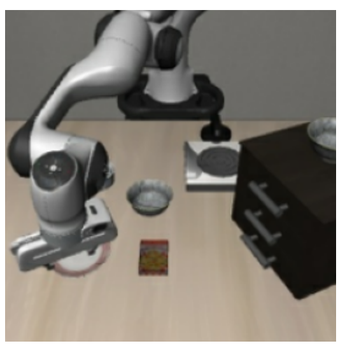
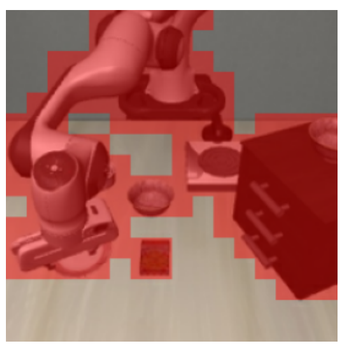
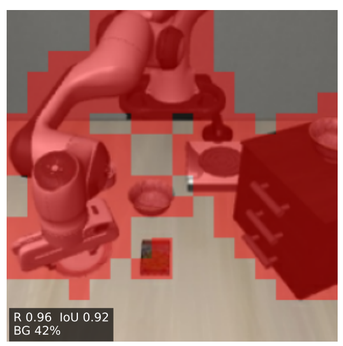
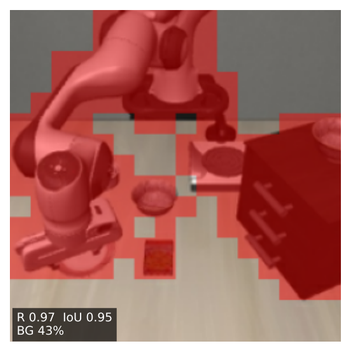
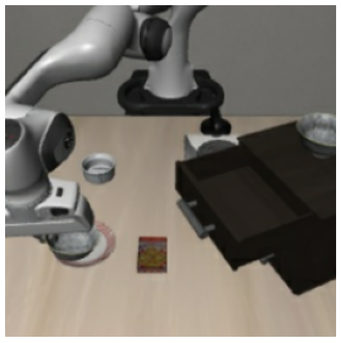
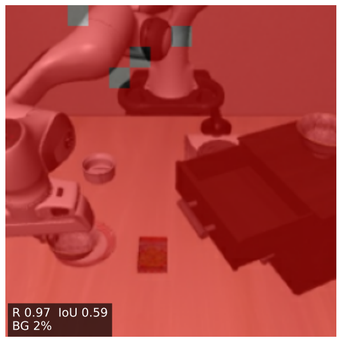
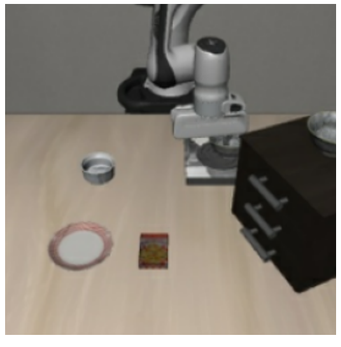
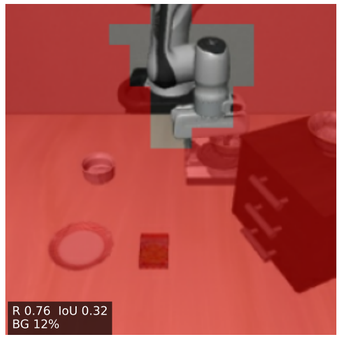
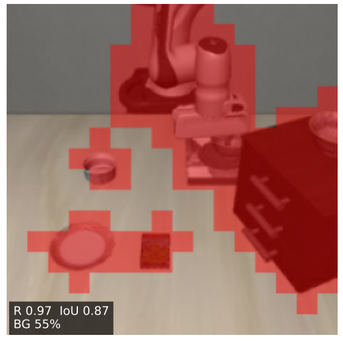
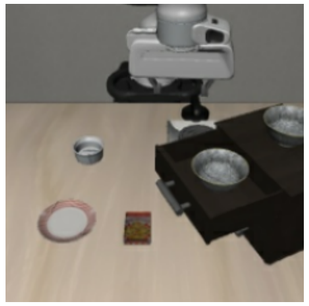
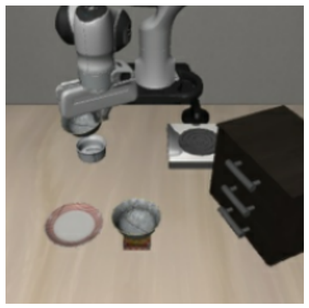
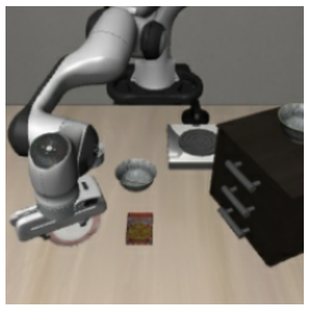
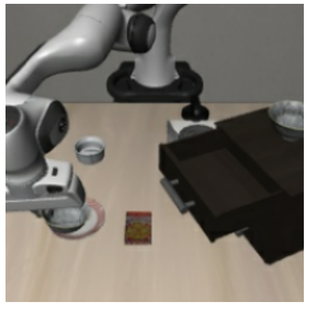
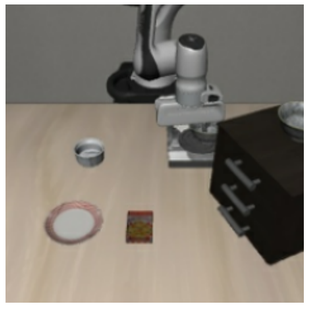
Track B · same observations across native-feature segmentation methods

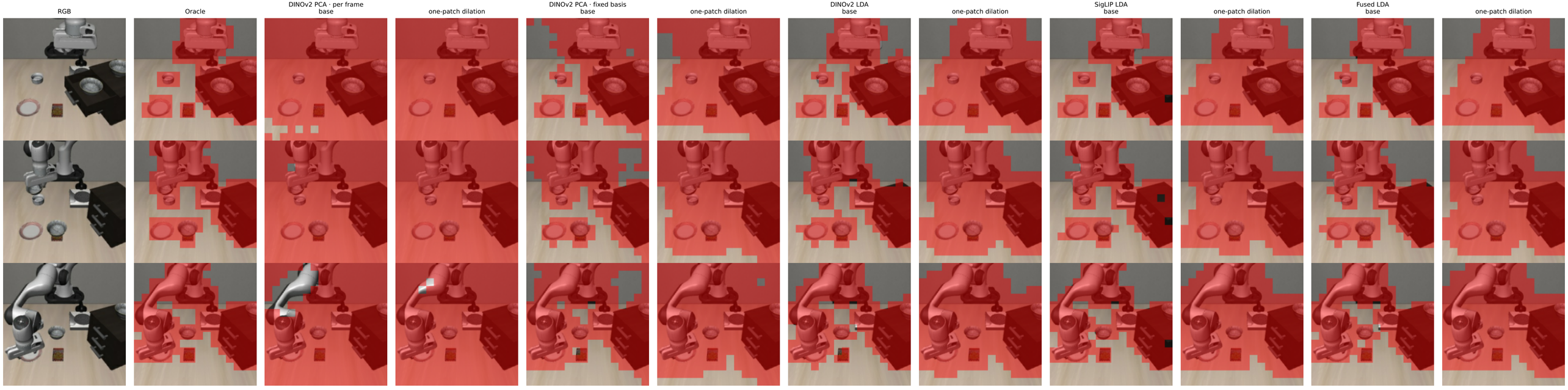
RGB observation	Oracle FG/BG	DINOv2 PCA · per frame	DINOv2 PCA · fixed basis	DINOv2 LDA	SigLIP LDA	Fused LDA
		 R 1.00 IoU 0.49 BG 3%	 R 0.99 IoU 0.72 BG 35%	 R 0.98 IoU 0.91 BG 50%	 R 0.98 IoU 0.98 BG 54%	 R 0.98 IoU 0.97 BG 52%
		 R 0.99 IoU 0.51 BG 0%	 R 0.98 IoU 0.75 BG 34%	 R 0.95 IoU 0.91 BG 48%	 R 0.95 IoU 0.94 BG 51%	 R 0.97 IoU 0.95 BG 49%
		 R 0.85 IoU 0.49 BG 9%	 R 0.98 IoU 0.76 BG 27%	 R 0.96 IoU 0.92 BG 42%	 R 0.95 IoU 0.95 BG 45%	 R 0.97 IoU 0.95 BG 43%
		 R 0.97 IoU 0.59 BG 2%	 R 0.98 IoU 0.79 BG 27%	 R 0.96 IoU 0.93 BG 40%	 R 0.96 IoU 0.95 BG 42%	 R 0.96 IoU 0.95 BG 42%
		 R 0.76 IoU 0.32 BG 12%	 R 0.98 IoU 0.64 BG 38%	 R 0.97 IoU 0.87 BG 55%	 R 0.99 IoU 0.95 BG 57%	 R 1.00 IoU 0.94 BG 56%

Token-level error maps · panels span splits; reported metrics use evaluation only

RGB observation	Oracle FG	DINOv2 PCA · per frame	DINOv2 PCA · fixed basis	DINOv2 LDA	SigLIP LDA	Fused LDA
		R 1.00 · IoU 0.49 · BG 3%	R 0.99 · IoU 0.72 · BG 35%	R 0.98 · IoU 0.91 · BG 50%	R 0.98 · IoU 0.98 · BG 54%	R 0.98 · IoU 0.97 · BG 52%
		R 0.99 · IoU 0.51 · BG 0%	R 0.98 · IoU 0.75 · BG 34%	R 0.95 · IoU 0.91 · BG 48%	R 0.95 · IoU 0.94 · BG 51%	R 0.97 · IoU 0.95 · BG 49%
		R 0.85 · IoU 0.49 · BG 9%	R 0.98 · IoU 0.76 · BG 27%	R 0.96 · IoU 0.92 · BG 42%	R 0.95 · IoU 0.95 · BG 45%	R 0.97 · IoU 0.95 · BG 43%
		R 0.97 · IoU 0.59 · BG 2%	R 0.98 · IoU 0.79 · BG 27%	R 0.96 · IoU 0.93 · BG 40%	R 0.96 · IoU 0.95 · BG 42%	R 0.96 · IoU 0.95 · BG 42%
		R 0.76 · IoU 0.32 · BG 12%	R 0.98 · IoU 0.64 · BG 38%	R 0.97 · IoU 0.87 · BG 55%	R 0.99 · IoU 0.95 · BG 57%	R 1.00 · IoU 0.94 · BG 56%

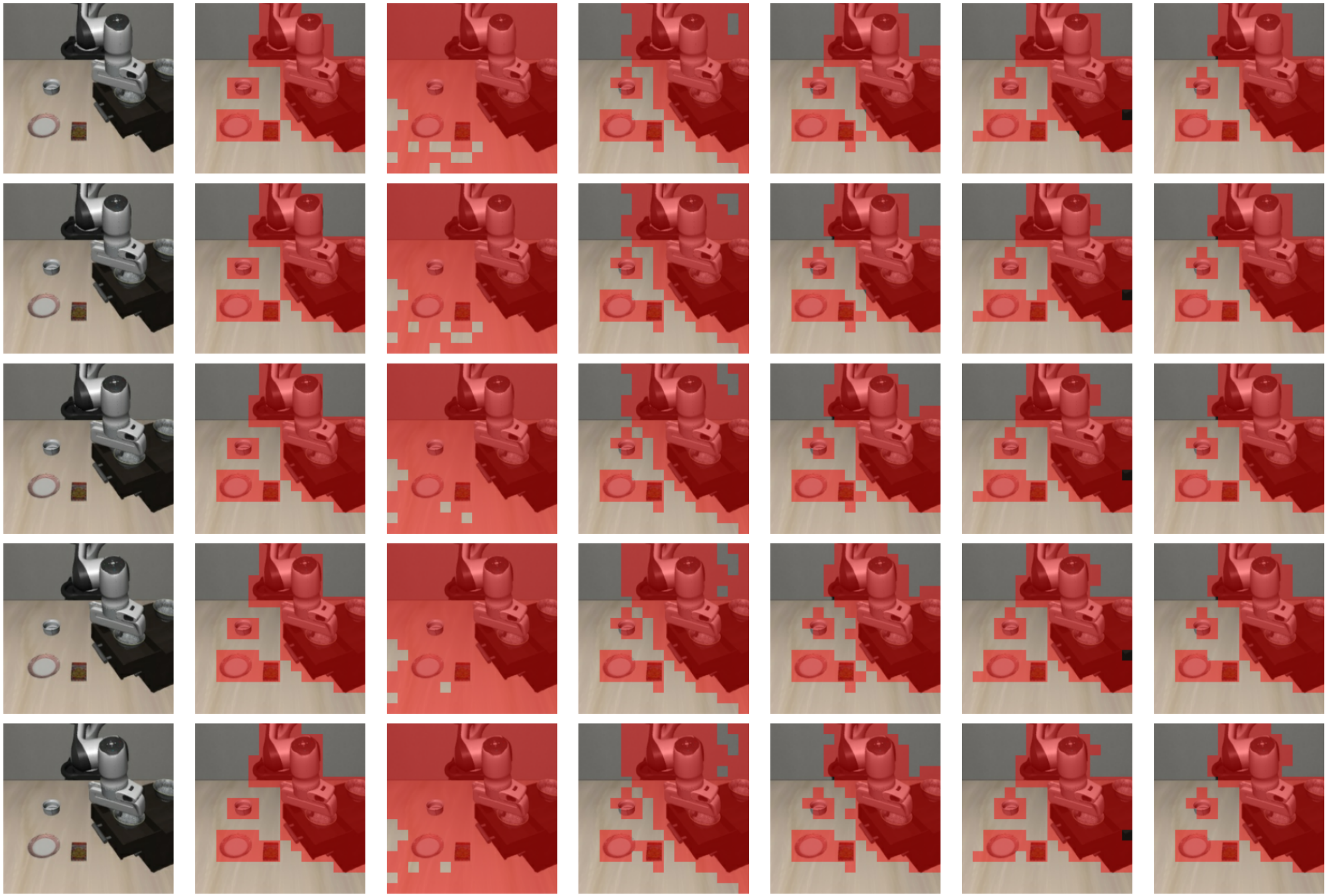
TP foreground FP foreground FN foreground TN background

Predicted foreground before and after 8-neighbor one-patch dilation



R5 temporal window · task 4 · task4_state2 · source step 60

RGB | Oracle | DINOv2 PCA · per frame | DINOv2 PCA · fixed basis | DINOv2 LDA | SigLIP LDA | Fused LDA



R5 temporal window · task 5 · task5_state2 · source step 40

RGB | Oracle | DINOv2 PCA · per frame | DINOv2 PCA · fixed basis | DINOv2 LDA | SigLIP LDA | Fused LDA

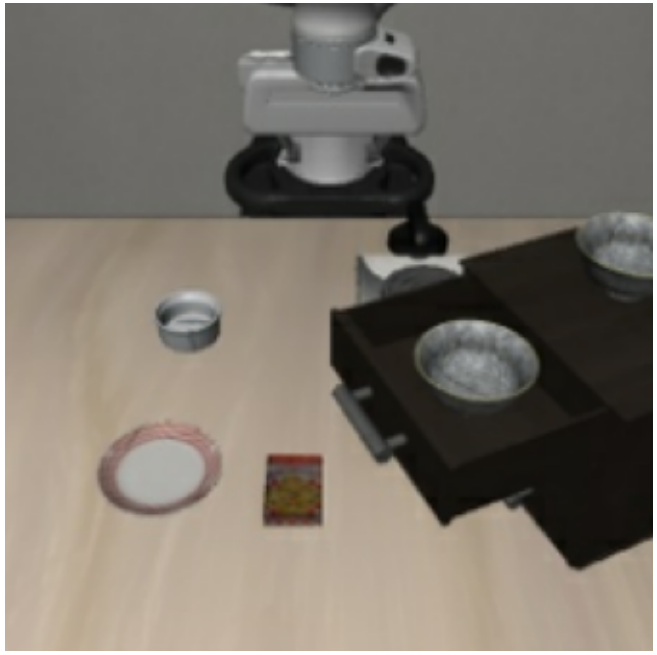


R5 temporal window · task 7 · task7_state2 · source step 110

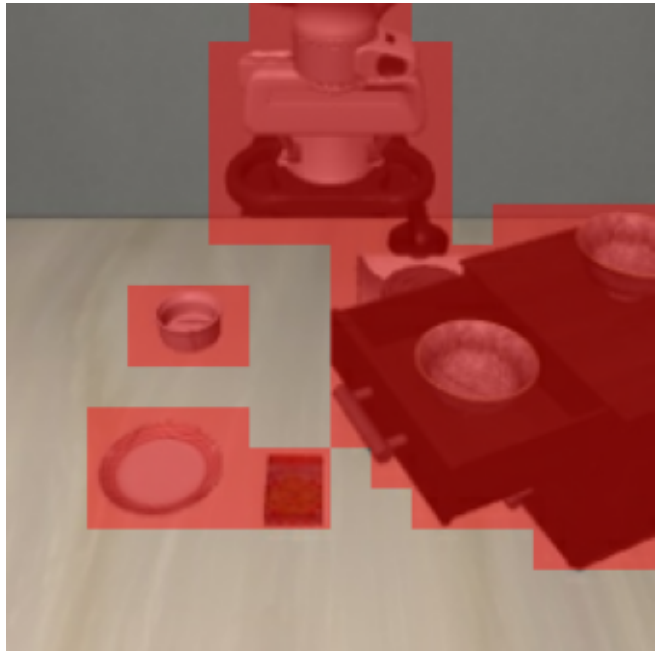


Spatial correspondence: 224×224 processor image → 16×16 OpenVLA visual tokens

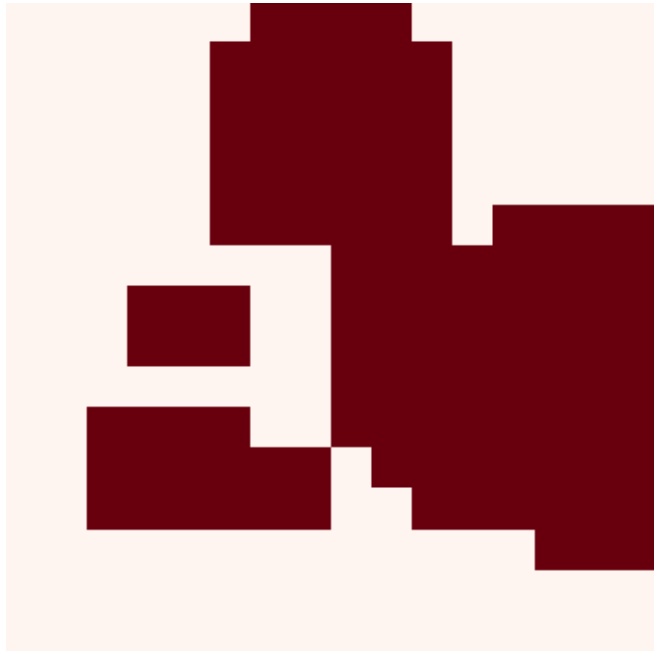
OpenVLA processor input

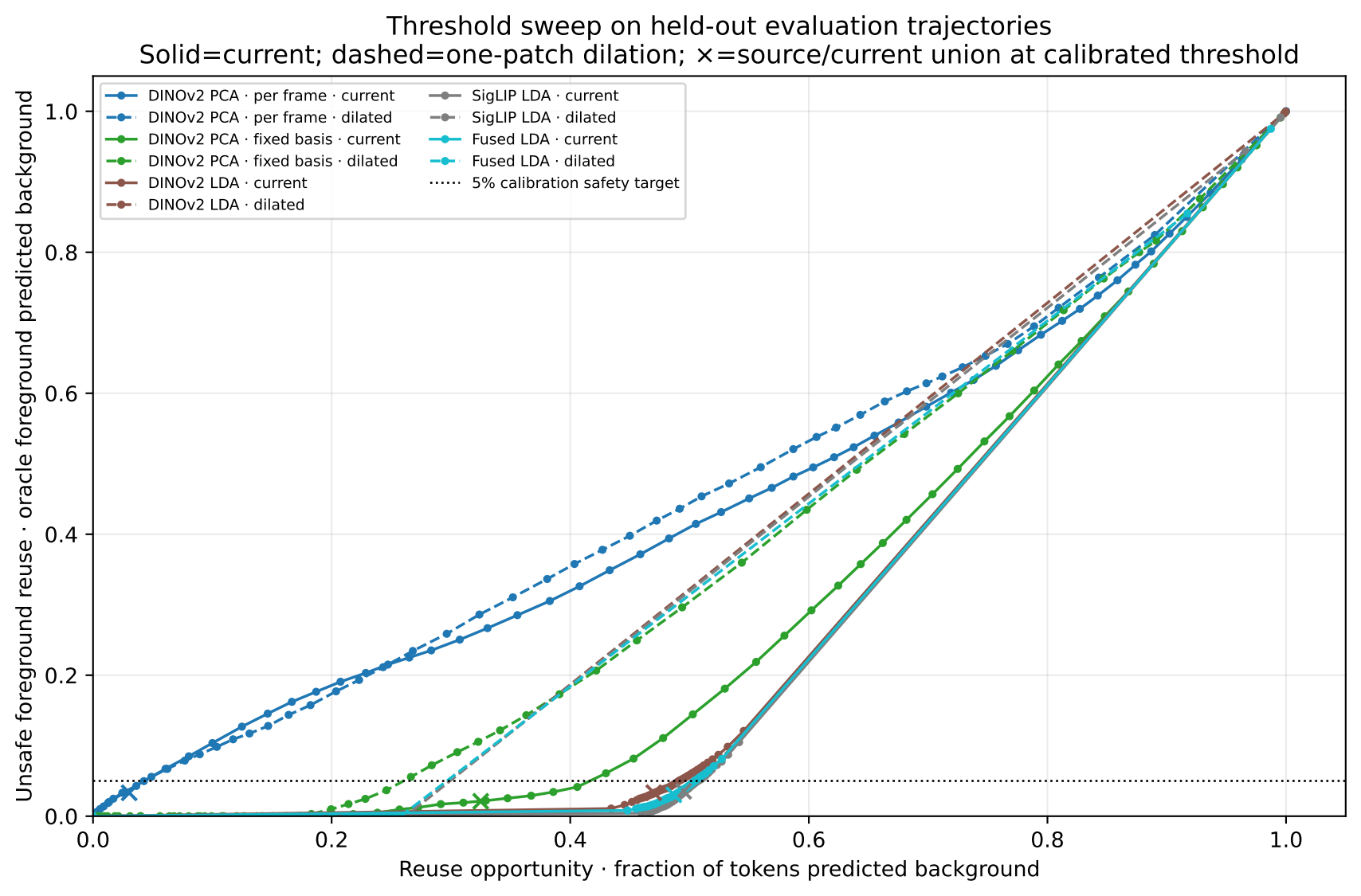


Oracle token mask on 16×16 patch grid



256 token foreground mask




Threshold sweep on held-out evaluation trajectories
Solid=current; dashed=one-patch dilation; ×=source/current union at calibrated threshold
DINOv2 PCA · per frame · current
DINOv2 PCA · per frame · dilated
DINOv2 PCA · fixed basis · current
DINOv2 PCA · fixed basis · dilated
DINOv2 LDA · current
DINOv2 LDA · dilated
SigLIP LDA · current
SigLIP LDA · dilated
Fused LDA · current
Fused LDA · dilated
5% calibration safety target
Unsafe foreground reuse · oracle foreground predicted background
Reuse opportunity · fraction of tokens predicted background
0.0
0.2
0.4
0.6
0.8
1.0